

AI / LLM Enterprise Application Case Study *

* Specific client sensitive information has been redacted and generalized



Impact



Impact:

This enterprise application helps mission-critical employees keep up with their tasking, making employees and leadership happy that important requests are being handled in a timely and efficient manner, reducing the backlog by 400%. This has improved employee morale and the call center is able to serve and protect the public as intended.

Project overview



The product:

Part of a suite of enterprise applications, this application handles incoming emailed requests for a call center. The customer suddenly had a 40-fold increase in demand, with tens of thousands of items being submitted weekly for a small team of employees to handle.



Project duration:

February 2025 – April 2025

Item Number	Priority	Category	Date Received
1234567890	URGENT	Category One Name	12/22/2023 03:58 EDT
0123456789	URGENT	Category Two Name	12/23/2023 10:59 EDT
2345678901	HIGH PRIORITY	Category Two Name	12/24/2023 12:04 EDT
3456789012	HIGH PRIORITY	Category One Name	12/26/2023 12:00 EDT
4567890123	HIGH PRIORITY	Category Three Name	12/27/2023 11:42 EDT
5678901234	LOW PRIORITY	Category Three Name	12/28/2023 12:00 EDT
6789012345	LOW PRIORITY	Category Three Name	12/29/2023 12:34 EDT
7890123456	LOW PRIORITY	Category Three Name	12/30/2023 11:32 EDT
8901234567	LOW PRIORITY	Category Four Name	12/31/2023 10:00 EDT
9012345678	LOW PRIORITY	Category Four Name	01/02/2024 11:00 EDT
1987654321	LOW PRIORITY	Category Two Name	01/03/2024 12:00 EDT

Project overview



The problem:

The small, overwhelmed staff could not keep up with the 40-fold increase in requests that they needed to handle. Time was of the essence, as they were getting more behind each day, so there was a tight deadline.



The goal:

To enable staff to keep up with new increased demands of requests by making the process as streamlined and easy as possible, with easier-to-understand steps and fewer steps overall.

Project overview



My role:

UI/UX Designer



Responsibilities:

Collaboration, problem-solving, user research, business requirements, wireframing, mockups, prototyping, user testing



My team:

UI/UX Designer, Developers, Business Analyst, Scrummaster, Project Manager

Understanding the user

- User research
- Analysis of User Pain Points
- Problem statements
- Ideation

User research: summary



I met with users twice to learn about their current pain points, as well as to understand their new processes now that their workload has greatly increased. Due to the fluid nature of their office workflows, these processes were continuing to evolve. The team continued to meet with the users to learn about new updates to their needs and requirements, based on this evolution. Possible issues about limitations were raised and confirmed.

Based upon this understanding, I took notes, made user flow diagrams, and made sketches of possible new workflows and user interfaces which would support these updates. Mockups were presented to users at 2 different times, and designs were iterated upon, following user feedback. Refinements resulted in even faster flows for some users, depending upon the category selected.

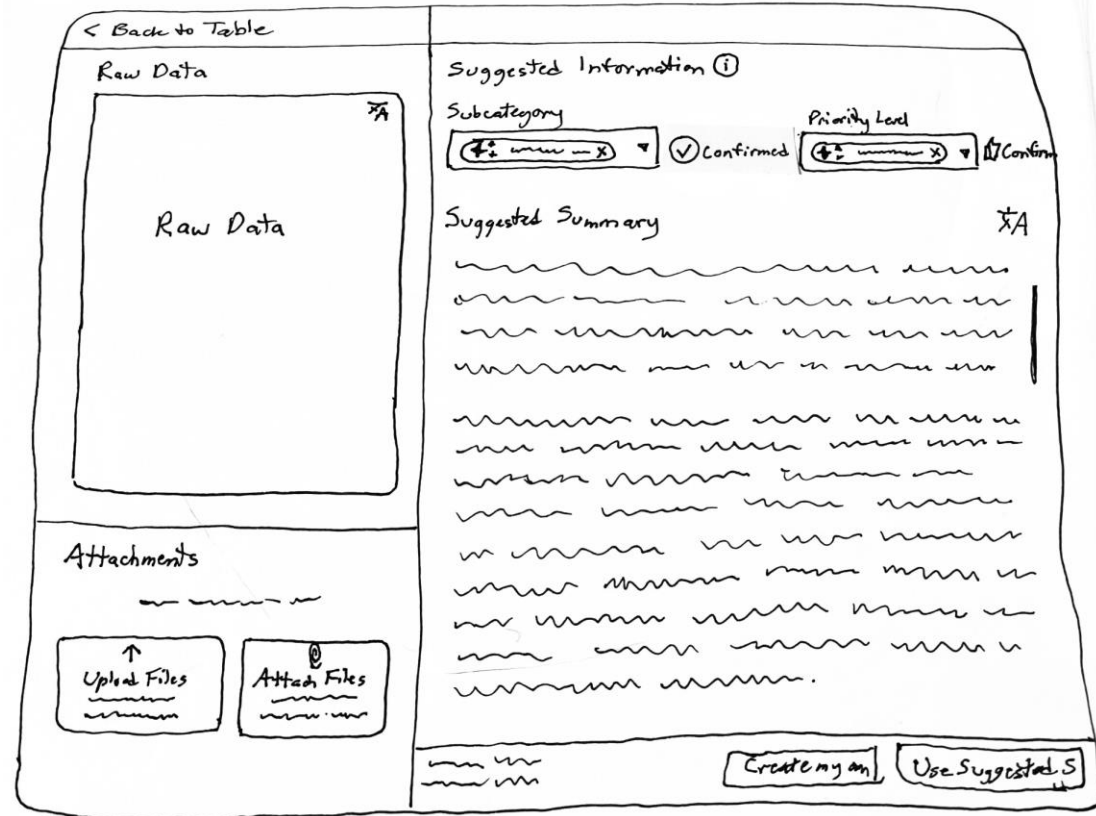
Research

The following questions needed to be researched and answered for this project:

- What is the information needed from the user during this process?
- Which of this information is mandatory and which is optional?
- What information is already known by the user at this stage in the process?
- Where does the information populate from?
- Is it possible for the system/LLM to 'guess' any of this information accurately?
- How can we provide the user with information that is known by the system at this point of the process?
- What do we do if the LLM-generated information is inaccurate? Do we allow users to manually override?
- How do we allow user to provide feedback on accuracy of LLM-provided information?
- Can the user undo this feedback? What if the user changes their mind?
- How can we explain to the user where the information was generated from? At what point is this necessary?
- What are the legal ramifications of using the LLM-generated information?

Wireframing

I explored the various screens needed for the app for the main user flows and how the layout could make this workflow intuitive for users and allow efficient streamlining of the steps.



Usability research: findings

1

Previously overwhelmed users appreciated the faster and more direct workflow.

2

Users liked being able to automate many of the fields, reducing manual typing. They also considered it important (and legally responsible) to be clearly aware of the difference between LLM-generated suggestions and manually input information.

3

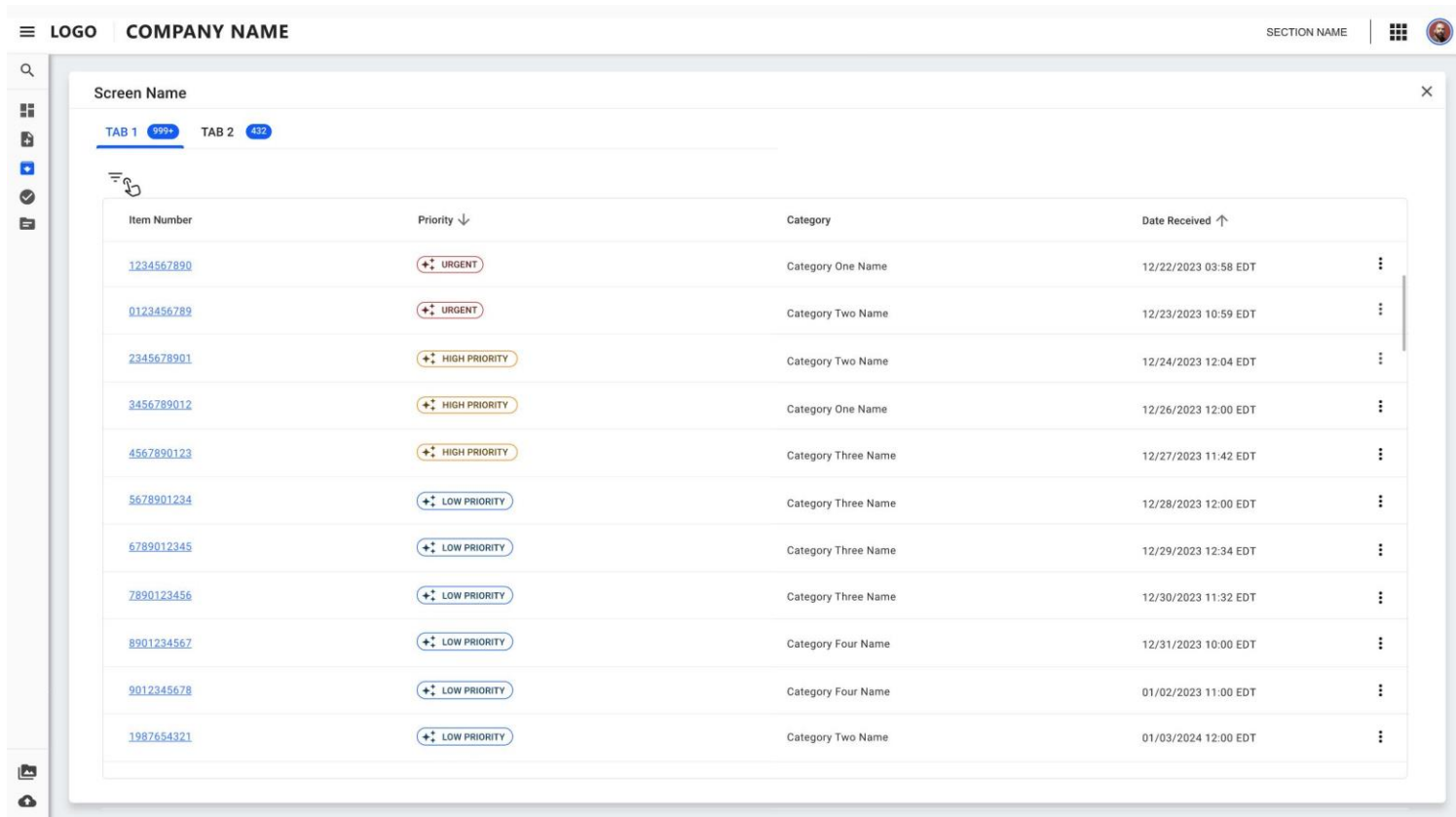
Users liked the simplified UI that clarified the steps they needed to take.

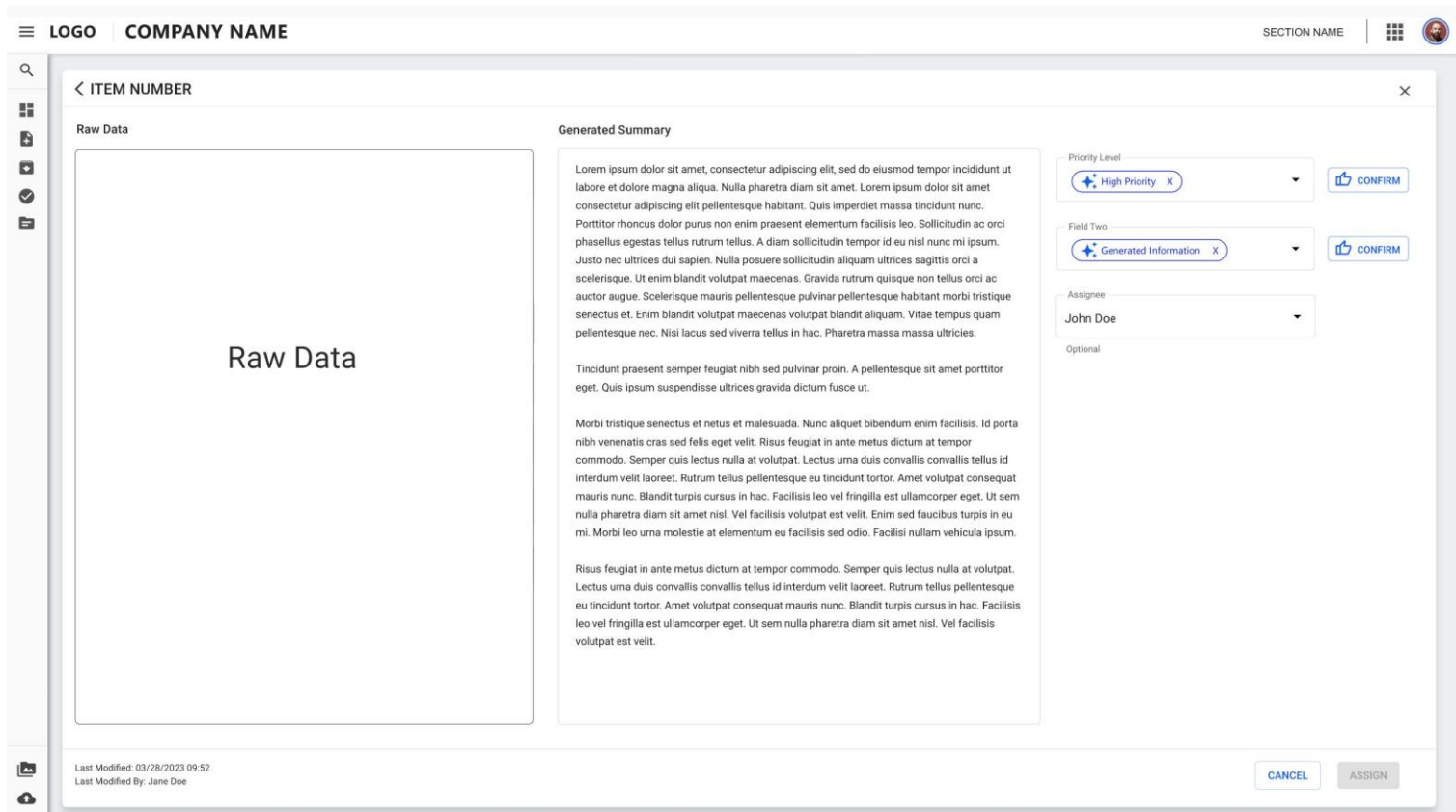
Refining the design

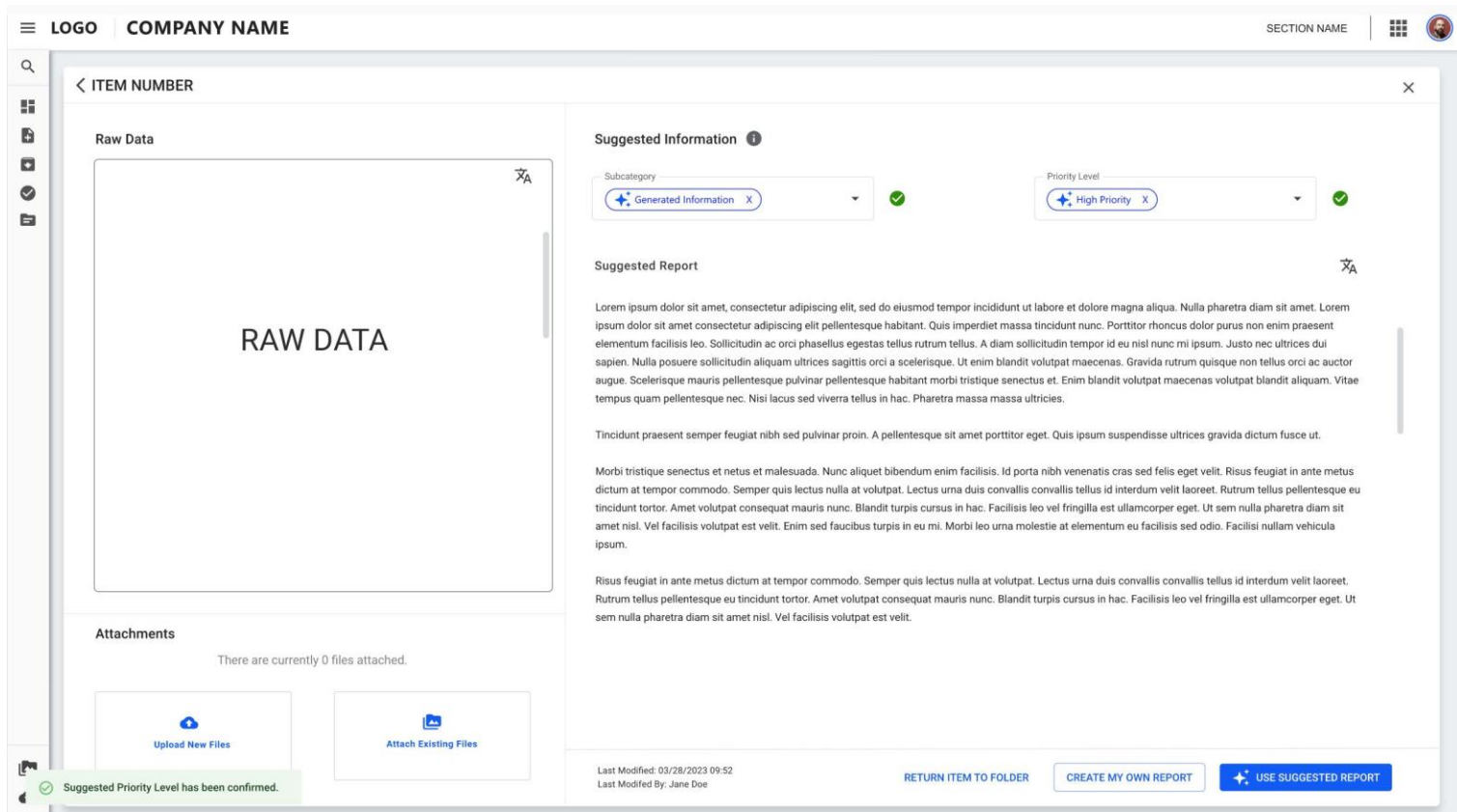
- Mockups
- High-fidelity prototype
- Accessibility

Mockups

High fidelity mockups were created to test with users and help the team understand the new workflow and layout.



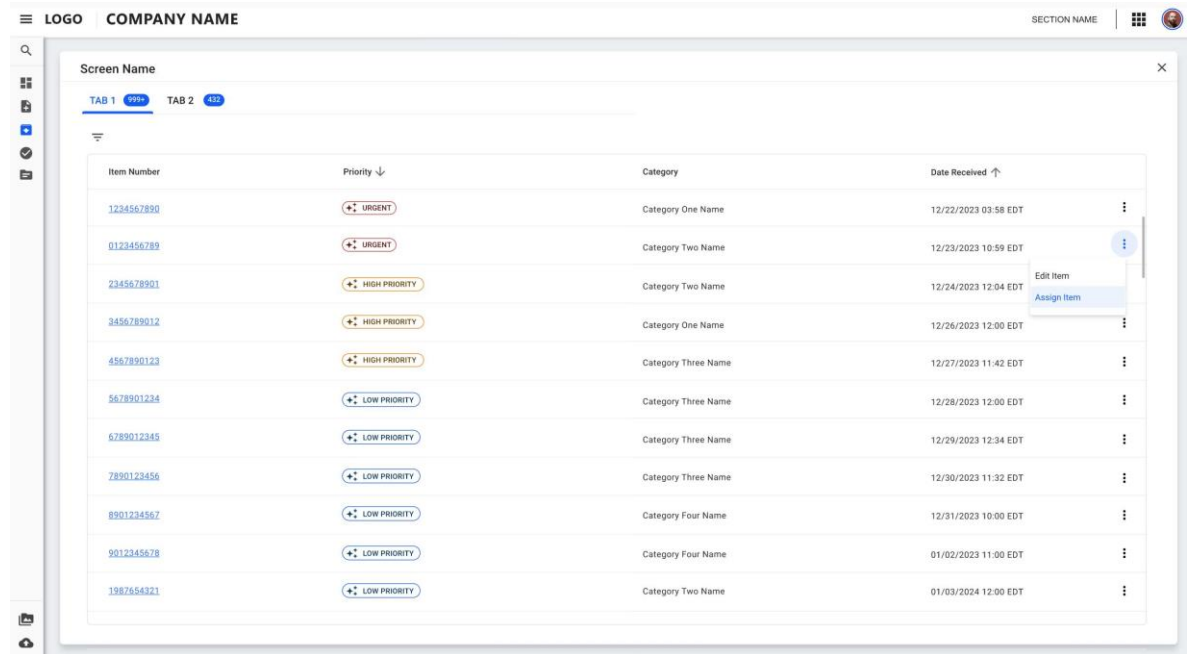




Prototype

I created an interactive clickable prototype so the team and users could understand how one would move through the screens for the various scenarios.

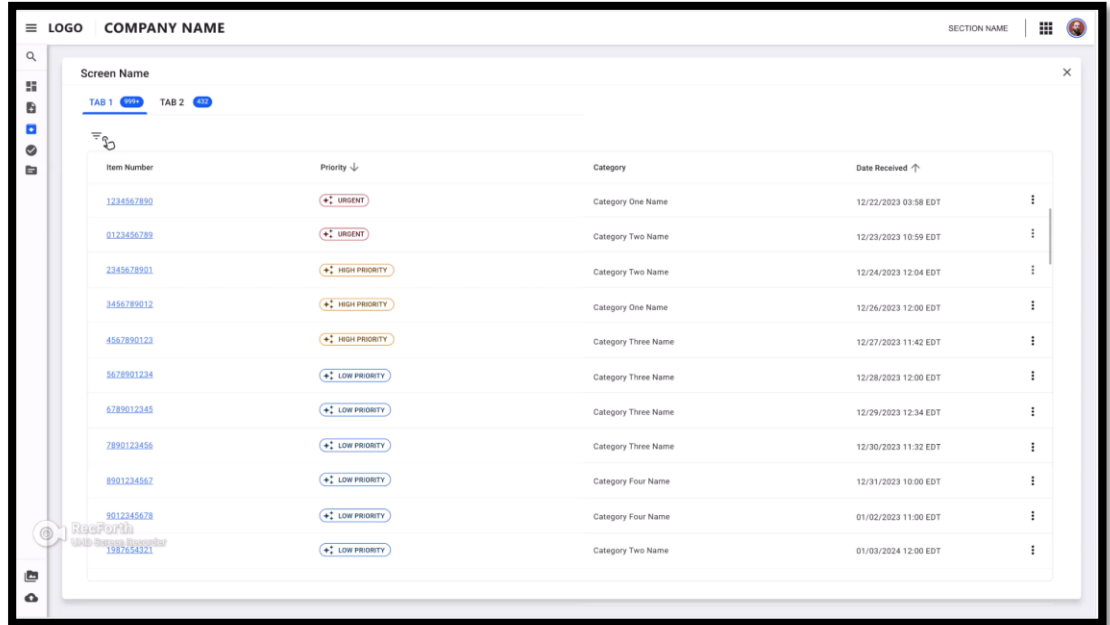
[View interactive prototype](#)



Prototype

The prototype includes various screens that the team and the client needed to visualize

Press Play button on video below to start



Accessibility considerations

1

Uses design system, which follows 508A accessibility guidelines for all components and is consistent with the rest of the platform suite. Design is adaptable with screen readers and colors have been tested for adequate contrast.

2

Includes feedback, clean layout, standardized language, and help options to aid user understanding of the new process and how to use AI, with clear distinctions between user-generated and system-generated content.

3

Allows user to accept AI suggestions or choose to perform steps manually, as desired, giving user control and flexibility.

Results and What I learned



Results: The organization is satisfied that critical tasks are being handled in a timely and efficient manner, **reducing the backlog by 400%.**

I learned that empathizing with and listening to users, while studying their pain points, can enable streamlining of business processes. This has a strong positive impact on their efficiency, results, and customer morale. Determining where and when AI automation can and should be applied is key. Determining important questions to research, answer, and understand will lead to a strong product. Design iterations may be necessary, but as long as each iteration brings the product, team, and the user closer together, this is a success story that can bring about real positive impact.

Let's connect!



Please contact me if you would like to review additional design work.

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